تجمع جازان الصحي
Jazan Health Cluster

King Fahad Central Hospital

IV Alteplase (tPA) Preparation & Administration Protocol

Acute Ischemic Stroke — Aligned with AHA/ASA Guidelines for Early Management of Acute Ischemic Stroke

1. Pre-Administration Verification (Time-Out)

Complete as a documented two-person independent double-check immediately before preparation and again before administration — standard for all high-alert medications.

☐ **Eligibility for IV thrombolysis reconfirmed: within approved time window from LKW, no exclusion criteria present, informed consent obtained per institutional policy**

☐ Non-contrast head CT reviewed — no hemorrhage, no contraindicating findings

☐ **Actual measured body weight used (never estimated or stated by patient/family) — confirm scale used and value: ______ kg**

☐ Blood pressure confirmed < 185/110 mmHg immediately before administration; treated per protocol if above threshold

☐ Point-of-care or laboratory glucose confirmed not severely abnormal (<50 or per institutional cutoff excludes stroke mimics)

☐ Relevant coagulation status reviewed (INR, platelet count, anticoagulant/antiplatelet history, DOAC last-dose timing) per protocol

☐ Correct drug (alteplase), correct concentration, correct total dose, and correct patient identifiers verified by two independent clinicians (e.g., RN + RN, RN + MD, or RN + Pharmacist)

☐ Emergency equipment and reversal-related supplies available at bedside (see Section 5)

2. Weight-Based Dosing

Parameter	Specification
Total dose	0.9 mg/kg actual body weight, MAXIMUM total dose 90 mg
Bolus component	10% of total calculated dose, given IV push over 1 minute
Infusion component	Remaining 90% of total calculated dose, infused IV over 60 minutes via programmable pump
Concentration after reconstitution	1 mg/mL

Worked example only — recalculate for each patient and verify independently: 70 kg patient → total dose 0.9 × 70 = 63 mg → bolus 6.3 mg (6.3 mL) over 1 min → infusion 56.7 mg (56.7 mL) over 60 min.

☐ **Dose calculation independently verified and documented by second clinician: Total dose ____ mg = Bolus ____ mg (____ mL) + Infusion ____ mg (____ mL)**

3. Preparation

☐ Obtain alteplase vial (50 mg or 100 mg) and matching manufacturer-supplied sterile water for injection diluent — do not substitute another diluent

☐ Reconstitute by directing the diluent stream into the vial per manufacturer instructions; gently swirl or slowly invert to dissolve — do NOT shake vigorously (causes excessive foaming)

☐ If foaming occurs, allow vial to stand undisturbed for several minutes until foam subsides

☐ Confirm final concentration is 1 mg/mL; inspect for particulate matter or discoloration — discard if present

☐ Withdraw calculated bolus dose into one syringe and calculated infusion dose into a second syringe or pump-compatible container, per institutional preparation workflow

☐ Label both syringes/containers clearly with drug name, concentration, total dose, patient identifiers, and time prepared

☐ Discard any unused reconstituted drug — do not save partial vials for later use

☐ Reconstituted solution is stable per manufacturer labeling (commonly up to 8 hours at room temperature or refrigerated) — verify current manufacturer prescribing information for storage/stability specifics at time of use

4. Administration

☐ **Administer via a dedicated IV line; do not infuse other medications through the same line**

☐ Do not administer anticoagulants or antiplatelet agents concurrently during the infusion window unless specifically ordered for another indication and reviewed by the stroke team

☐ Give bolus dose (10% of total) IV push over 1 minute

☐ Immediately follow with infusion of remaining 90% of total dose via programmable infusion pump over 60 minutes

☐ Second clinician independently verifies pump programming (drug, concentration, dose, rate, duration) before infusion starts

☐ **Document exact bolus administration time — this is the institutional "door-to-needle" / "needle time" quality metric**

☐ Flush line per institutional policy after infusion completes

☐ Avoid arterial punctures, IM injections, urinary catheter placement, and nasogastric tube placement during and shortly after infusion unless essential — minimize bleeding risk

5. Monitoring During & After Infusion

☐ Neuro checks + vital signs: q15min × 2h (starting at bolus time), then q30min × 6h, then q1h × 16h — per institutional post-tPA protocol

☐ Blood pressure maintained < 180/105 mmHg throughout monitoring period; treat per protocol and notify physician if exceeded

☐ No antithrombotic (antiplatelet/anticoagulant) agents for 24 hours post-infusion, and only after repeat imaging excludes hemorrhage

☐ Repeat non-contrast head CT at 24 hours, or immediately if any neurologic decline

☐ Avoid or delay central line placement, arterial line placement, and NG tube insertion during the first 24 hours unless essential

Full post-thrombolysis monitoring detail (diet, DVT prophylaxis, positioning, etc.) follows the institution's Acute Ischemic Stroke admission order set.

6. Management of Suspected Complications

6a. Orolingual Angioedema

☐ Stop infusion immediately; assess and secure airway — have emergent airway equipment available

☐ Administer antihistamine (e.g., diphenhydramine) and corticosteroid per institutional anaphylaxis/angioedema protocol

☐ Administer epinephrine if severe/rapidly progressive airway compromise, per protocol

☐ Notify physician immediately; consider ENT/anesthesia for airway management if progressive

6b. Suspected Symptomatic Intracranial Hemorrhage

☐ **Stop infusion immediately if still running**

☐ Obtain emergent non-contrast head CT

☐ Obtain STAT labs: CBC, PT/INR, aPTT, fibrinogen, type and crossmatch

☐ Notify physician/stroke team and neurosurgery immediately

☐ Consider reversal/hemostatic measures per institutional hemorrhage-reversal protocol (e.g., cryoprecipitate, antifibrinolytic agents) — dosing and agent selection per current institutional protocol and hematology/neurosurgery input

☐ Manage blood pressure and ICP per neurosurgical/neurocritical care direction

6c. Systemic Bleeding (non-intracranial)

- ☐ Apply local hemostatic measures (direct pressure) for accessible bleeding sites
- ☐ Assess severity; stop infusion if severe/hemodynamically significant
- ☐ Obtain CBC, coagulation studies, type and crossmatch as indicated
- ☐ Supportive transfusion and reversal measures per institutional protocol if significant blood loss

7. Documentation

Patient Name: ______________________ **MRN:** ___________________	**Last Known Well (LKW):** __________ **Time drug ordered:** __________
Measured Actual Body Weight: ______ kg	**Time bolus given ("needle time"):** __________

- ☐ Time IV access obtained
- ☐ Time drug prepared and verified (both clinicians documented by name)
- ☐ Time bolus started and completed
- ☐ Time infusion started and completed
- ☐ Any adverse events, interruptions, or complications and actions taken
- ☐ Door-to-needle time calculated and reported to stroke registry/Get With The Guidelines–Stroke

Physician Order & Verification

Ordering Physician Signature: ________________________ Date/Time: _________________

Preparer (RN/Pharmacist): ________________________ Independent Verifier: _________________

References: 2019 AHA/ASA Guidelines for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates; current manufacturer prescribing information for alteplase; AHA/ASA Get With The Guidelines–Stroke door-to-needle measures; The Joint Commission Primary/Comprehensive Stroke Center certification and high-alert medication requirements; institutional Pharmacy & Therapeutics thrombolytic policy. This template must be re-verified against the current prescribing information and guideline version at the time of institutional adoption — drug concentrations, stability windows, and reversal protocols are subject to change.